

we can ignore the difference between the Superclass and Subclass objects.

An object of a Subclass can be treated as an object of its Superclass. What would stop this from being true? Is there any part of the Superclass that is missing from the Subclass? Well no it's all there. Certainly if we are asking a Subclass to fill in for its parent, we throw away the extra stuff that the Subclass added, but—because a Subclass “is an” example of its Superclass—we can “treat the Subclass as an example of the Superclass.” And, if we only want to call the methods from the Superclass and access the data variables that come from the Superclass—what's the difference?

Doing this might be interesting. As long as we treat the reference like it only refers to the Superclass, we can handle a Subclass object from a Superclass “reference.” As long as we only call the methods that exist in the Superclass, this will work fine.

```
public class Employee
{
    String name;
    String SSN;
    public double calc_pay()
    {
        return 0.0;
    }
}

public class HourlyEmployee
extends Employee
{
    double hourly_rate;
    int hours;
    public double calc_pay()
    {
        return hours * hourly_rate;
    }
}

public class SalariedEmployee
extends Employee
{
```